Stage 1: Multitask Preference Probing

Prompt: ⟨Prmopt 1⟩. Given the user's Category sequence: {(Pet Services, Sun, 10am), (Movie Theaters, Sun, 11am), (Coffee Shop, Sun, 2pm), (Grocery, Sun, 4pm), (Bars, Sun, 8pm)} what is the user's categorical transition preference? Considering: what are the 'category pairs' the user usually visits consecutively? (format:category-category,...)

Output: ***{Pet Services-Movie Theaters, Movie Theaters-Coffee Shop, Coffee Shop-Grocery, Grocery-Bars}.***

Prompt: What is the user's categorical temporal preference? Considering: what are the 'categories' the user visits at a certain time (day/ hour)? (format:{time: [categories]})

Output: ***{morning: [Pet Services, Movie Theaters], afternoon: [Coffee Shop, Grocery], evening: [Bars]}.***

Stage 2: Dynamic Preference Self-Reflection

Prompt: The user has visited categories {(Coffee Shop, Mon, 8am), (Pet Services, Mon, 10am). Now is Mon at 12pm, based on the user's categorical transition preference and categorical temporal preference, predict users' next most likely visiting 'Category'. (format: Category)

Output: ***Movie Theatres.***

Prompt: The user actually visited {Restaurants}. Based on this, what is the new insight you can get for the user's categorical transition preference? Generate the updated categorical transition preference. (format:{category-category,...})

Output: ***{Pet Services-Movie Theaters, Movie Theaters-Coffee Shop, Coffee Shop-Grocery, Grocery-Bars, Coffee Shop-Pet Services, Pet Services-Restaurants}.***

Stage 3: Neighbor Preference Summarization

Prompt: The users' geographical neighbors' categorical transition preferences are {Restuarants-Bars, Pet Services-Coffee Shop}; the users' semantic neighbors' categorical transition preferences are {Coffee Shop-Pet Services, Pet Services-Restaurants}; the users' social neighbors' categorical transition preferences are {Department Store-Clothing, Restaurants-Bars, Grocery-Bakeries}. Summarize the neighbors' categorical transition preferences by considering their commonalities. (format: {category-category,...}).

Output: ***{Restaurants-Bars, Pet Services-Coffee Shop, Pet Services-Restaurants, Department Store-Clothing, Grocery-Bakeries}.***

Stage 4: Next POI Recommendation

Prompt: Now is {Mon} at {12pm}, based on the users' current category sequence {Coffee Shop, Mon, 8am), (Pet-Services, Mon, 10am)}, his own categorical transition preference and categorical temporal preference, and his neighbors' categorical transition preference and categorical temporal preference, predict the user's next most likely visiting 'category'. (format: category)

Output: ***Restaurants.***

⟨Next Region and Distance Prediction⟩

Prompt: Candidate POIs: (format:(Name, Category, Region, Distance)): {(The Melt, Restaurants, r3, 1km), (Crescent Ballroom, Bar, r3, 2km), (Pizzeria Bianco, Restaurants, r1, 12km), ...} Given users' current check-in sequence {(Espresso Bar, Coffee Shop, r3, Mon, 8am), (Petbar Phoenix, Pet Services, r3, Mon, 8am)}, recommend 10 POIs from the candidate set, considering his next likely visiting category {Restaurants}, region{r3}, and distance{2km}. State the reason for each recommendation and rank the importance of category, region, and distance preferences. (format: {POI: reason; [importance ranking]})}

Output: ***The Melt: Matches next likely category 'Restaurants', in preferred region 'r3', and within 2 km distance; [Category > Region > Distance].***